HDOC DAY – 43

ALGORITHM:

1. Start.
2. Display the menu:
    - Find the longest word in a sentence.
    - Check whether one string is a rotation of another.
    - Exit.
3. Read the user's choice.
4. If choice is 1:
    - Read a sentence.
    - Separate/check each word of the sentence.
    - Find the length of each word.
    - Compare it with the maximum length found so far.
    - Store the word if its length is greater.
    - Display the longest word.
5. If choice is 2:
    - Read two strings.
    - Check whether their lengths are equal.
    - If lengths are different, they are not rotations.
    - Otherwise, concatenate the first string with itself.
    - Search for the second string in the concatenated string.
    - If found, the strings are rotations; otherwise, they are not.
6. If choice is 3, exit the program.
7. For any other choice, display an invalid-choice message.
8. Repeat the menu until Exit is selected.
9. Stop.

C PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>

// Function to find the longest word
void longestWord()
{
   char sentence[200];
   char word[50], longest[50];
   int i = 0, j = 0;
   int maxLength = 0;

   printf("Enter a sentence: ");
   getchar();
   fgets(sentence, sizeof(sentence), stdin);

   while (1)
   {
      // Store characters of one word
      if (sentence[i] != ' ' &&
         sentence[i] != '\n' &&
         sentence[i] != '\0')
      {
         word[j] = sentence[i];
         j++;
      }
      else
      {
         // End the current word
         word[j] = '\0';
```

```
        if (j > maxLength)
        {
            maxLength = j;
            strcpy(longest, word);
        }

        j = 0;

        if (sentence[i] == '\0')
            break;
      }

      i++;
   }

   printf("Longest word: %s\n", longest);
   printf("Length: %d\n", maxLength);
}


// Function to check string rotation
void checkRotation()
{
   char str1[100], str2[100];
   char temp[200];

   printf("Enter first string: ");
```

```
    scanf("%99s", str1);

    printf("Enter second string: ");
    scanf("%99s", str2);

    // Rotations must have the same length
    if (strlen(str1) != strlen(str2))
    {
        printf("The strings are NOT rotations of each other.\n");
        return;
    }

    // Join str1 with itself
    strcpy(temp, str1);
    strcat(temp, str1);

    // Search str2 inside str1 + str1
    if (strstr(temp, str2) != NULL)
        printf("The strings are rotations of each other.\n");
    else
        printf("The strings are NOT rotations of each other.\n");
}


int main()
{
    int choice;
```

```
do
{
   printf("\n----- MENU -----\n");
   printf("1. Find longest word in a sentence\n");
   printf("2. Check if one string is a rotation of another\n");
   printf("3. Exit\n");

   printf("Enter your choice: ");
   scanf("%d", &choice);

   switch (choice)
   {
      case 1:
         longestWord();
         break;

      case 2:
         checkRotation();
         break;

      case 3:
         printf("Program terminated.\n");
         break;

      default:
         printf("Invalid choice! Please try again.\n");
   }
```

```
    } while (choice != 3);


    return 0;
}
```

TEST CASE TABLE:

Test Case	Choice	Input	Expected Output
1	1	I love programming	Longest word: programming
2	1	C is very easy	Longest word: very
3	2	ABCD, CDAB	Strings are rotations
4	2	hello, llohe	Strings are rotations
5	2	hello, world	Strings are NOT rotations
6	3	—	Program terminated